

Paolo De Marinis

MSc in Mathematics | ML Foundations | Quantitative Analysis & Agentic AI Workflows

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Professional Profile

MSc graduate in Mathematics (106/110) with a strong quantitative background and university training in algorithms and numerical methods. My Master's thesis, written in English, is a largely self-contained theoretical reconstruction of zk-STARK protocols with formal contributions on Algebraic Intermediate Representations (AIR) and their optimization. Technical and scientific project work includes an algorithmic contribution to reversible 2D data compression, quantitative analysis, Linux/ACPI diagnostics and C programming. I have studied core supervised and unsupervised machine-learning methods and am continuing structured self-study in machine learning and AI. Python experience is project-based, mainly involving pipeline structure and design, code reading and review, targeted modifications, testing and validation; my direct programming and scientific-computing experience is stronger in C and MATLAB. I use agentic AI workflows for implementation support while retaining responsibility for problem framing, objectives and constraints, review, testing, validation and final acceptance.

Education

MSc in Mathematics (LM-40) - 106/110

2023-2025

University of Insubria, Italy

Thesis: *Fast Fourier Transform in zk-STARK Protocols: Interpolation, Encoding and Proximity Proofs*, written in English. Largely self-contained theoretical reconstruction of zk-STARKs from first principles. Theoretical contributions include a formal derivation from a finite-state machine to a Computational Integrity statement and Algebraic Intermediate Representation (AIR), with proof of equivalence between AIR satisfiability and CI-statement validity; a general AIR definition unifying formulations in the literature; and an optimization framework based on two preorders for valid and probabilistically sound extensions. The thesis also reconstructs the STARK protocol through Reed-Solomon encoding, DEEP and FRI, with complexity and soundness analysis.

Technical Skills

- **Machine learning and AI study:** already studied supervised and unsupervised learning, including SVM, decision trees, k-NN, k-means, GMM/EM and PCA, as well as knowledge representation, logic and knowledge bases. Continuing structured study includes generative methods, deep learning, embeddings, transformers, foundation models and LLMs; semi-supervised, self-supervised and reinforcement learning; big data and data mining.
- **Python project work:** pipeline structure and design, code reading/review, targeted modifications, testing and validation for a reversible pipeline mapping 2D points to Hilbert indices, ordering the indices and applying delta encoding to the ordered sequence, with inverse decoding from deltas to indices and from indices back to points. The implementation used [hilbertcurve](#), with targeted ChatGPT support for some functions.
- **Agentic AI workflows:** definition of initial objectives and constraints, context preparation, diff review, test execution and monitoring, result validation and final acceptance. Current agentic development primarily uses Codex and Claude Code/Cowork; technical analysis primarily uses ChatGPT and Claude. Additional experience includes Gemini, Cursor and Google Antigravity.
- **Direct programming and scientific computing:** C and MATLAB. University C++ work was mainly procedural and C-style; object-oriented C++ was limited to a single program for the pass/fail Programming Languages exam. GNU Octave was used directly in the Numerical Analysis exam; Maple was learned independently and used for orbital-transfer and rendezvous calculations presented in the Space Mechanics examination seminar. University training in algorithms and data structures.
- **Technical environment and tools:** Linux daily since 2012; practical use of Git/GitHub and GitHub Actions; advanced LaTeX for academic and technical documentation. Limited direct project experience with HTML and SQL on a PrestaShop e-commerce site.

Selected Projects

OMEN ACPI Toolkit - Linux/ACPI diagnostics with agentic workflow

Jul 2026-present (maintenance)

Repository | [v2.4.0 release](#) (14 Aug 2026); unaffiliated with HP.

- Used logs and controlled experiments to distinguish incomplete S5 shutdown, which left the dGPU powered, from a separate WQBZ ACPI error on the sole physically validated configuration: HP OMEN MAX 16-ap0006sl, board 8E35, BIOS F.13.
- I own hardware evidence, problem framing, objectives/constraints, safety decisions, diff review, log/regression analysis, real-hardware validation and releases. Codex made substantial contributions to Bash/Python implementation, automated tests and documentation. No independent specialist or line-by-line audit is claimed.
- The toolkit uses an existing firmware path for an S5-scoped override and includes local table capture, fail-closed guards, round-trip verification, recovery and separate Limine entries preserving stock boot paths. v2.4.0 improves packaging, documentation and release engineering; ACPI transformers, verifiers, patches and fixtures are unchanged from the v2.3.0 validated baseline.

Geospatial data compression - algorithmic contribution

2023

Voluntary, unpaid scientific collaboration; non-author contribution acknowledged in [Liverotti et al., International Journal for Numerical Methods in Engineering \(2025\)](#). Contributed to the design and Python implementation of a reversible 2D pipeline: points to Hilbert indices, ordering, delta encoding, inverse decoding and reconstruction. The implementation used [hilbertcurve](#), with targeted ChatGPT support for some functions; I personally verified round-trip reversibility on real Earth Observation data. Also proposed Huffman coding for brightness values and contributed to the triangulation-reconstruction approach.

RIVES cartridges - C games and procedural geometry

2024; GitHub repositories 2026

Original production and Jam (Oct–Dec 2024): conceived and published *Bomb Flip* first and then *Slither Slide* in C. For *Bomb Flip*, I designed the specific rules and wrote most original gameplay code; Cursor mainly supported animation and implementation details. For *Slither Slide*, I designed mechanics, worked on parts of the code, integrated/validated the result and selected/configured the closed uniform cubic B-spline; assistance was greater. *Slither Slide* placed 2nd of 7 entries in RIVES Jam #3. GitHub repository phase (Aug 2026): I created and now maintain the public *Bomb Flip* (v1.0.0) and *Slither Slide* (v1.0.0) repositories from the available post-jam versions, not exact 2024 source snapshots. I integrated provenance, documentation, tests, CI, cartridges and checksums; Codex assisted with maintenance, refactoring, tests and documentation, which I reviewed and validated.

Coffee extraction - quantitative analysis in MATLAB

Aug 2019-Feb 2020

Independent voluntary and unpaid project requested by Caffè dei Cavalieri, Pisa. Designed the analysis and calculations, measured total dissolved solids (TDS), built least-squares extraction curves in MATLAB and interpreted the results for a mixed infusion/percolation process. The analysis was presented by Matteo Cecconi at the 2020 Italian Brewers Cup, where he placed 2nd.

Additional Education

Master's studies in Mathematics - 75 ECTS completed before transfer

2016-2023

University of Pisa, Italy

Completed 75 ECTS through October 2020; enrollment was subsequently maintained during a study pause before transfer to the University of Insubria in 2023. Of the 75 ECTS, 71 were recognized on transfer. Coursework included Numerical Analysis, Scientific Computing, Dynamical Systems, Orbital Determination, Celestial Mechanics, Space Mechanics and Planetary Systems. Academic work included direct use of GNU Octave in the Numerical Analysis exam; use of independently learned Maple for orbital-transfer and rendezvous calculations presented in the Space Mechanics examination seminar; preparation of an independent 47-slide seminar on Gauss, Laplace and Moschetti methods for preliminary orbit determination; and an individual study of the Yarkovsky effect on asteroids for the Planetary Systems exam.

BSc in Mathematics (L-35)

2012-2016

Sapienza University of Rome, Italy

University coursework included procedural, C-style programming in C++, algorithms and data structures, and MATLAB used in the Numerical Analysis exam and BSc thesis. Thesis on Brouwer's fixed-point theorem and periodic solutions of ODEs, including numerical phase portraits in MATLAB.

Professional Experience

Computer Science Teacher (A-41) & Digital Trainer

Sep 2023-Aug 2024

IIS Leonardo da Vinci and Istituto Professionale Crotto Caurga, Italy

Taught computer science across technical and vocational programs, including C/C++ programming, spreadsheets and digital tools. Served on the digital team and delivered 16 hours of adult training on Google Workspace, plus EIPASS-related courses. Managed parallel planning across multiple classes and institutions, adapting communication and priorities to different contexts.

Mathematics Teacher

Sep 2022-Aug 2023

IIS Agrario Antonio della Lucia, Feltre, Italy

Taught mathematics across technical, vocational and professional-training programs.

Additional Technical Experience

PrestaShop e-commerce customization - voluntary, unpaid

May 2016-2022

New Zealand Sport S.r.l.s.

Set up and maintained aorakiski.com from an existing platform and theme; directly performed targeted HTML customization, module configuration and SQL queries and made targeted changes to data and search-related settings.

Languages

Italian: native. English: B2; used to write the MSc thesis and technical documentation and to read scientific literature.